B. Tech.(Computer Engineering) Vth Semester
1st Sessional 2020-21
AUTOMATA THEORY (CEN-502)

Exam Time: 1 Hour **MM:15**
Submission Time : 15 Mins
Note: Attempt all questions. **Date : 20/10/21**

Mention your Name and Roll No on each page of Answer sheets
Answer sheets MUST be submitted within the stipulated time.
Submit answer sheets on Official Google Classroom of Automata theory.

Q No.	CO Mapped
1	1
2	2
3	1
4	1
5	2

Q. No	Question	Marks
1.	Construct a DFA for a Language over $\sum = (0,1)$, such that each string of this language has the last three elements same. **Verify your answer with suitable examples.**	2
2	With the help of Arden's theorem and representing each step involved, deduce the Regular Expression for the DFA asked in Question 1 above.	2
3.	Give suitable reasons to answer the following a) How many states will be there in a DFA which accepts strings with ***'x'*** no of 0's, and ***'y'*** no of 1's. b) How many states will be there in a Moore machine equivalent to a '***k***' states and ***'p'*** output Mealy machine?	3
4	Using suitable reasons, comment on the VALIDITY of each of the following statements. a) If Language L_1 and L_2 are Regular, then $\mathbf{L_1 \cup L_2}$ is also Regular. b) If Language L_1 and L_2 are Regular, then $\mathbf{L_1.L_2}$ is also Regular. c) If Language L_1 is regular and Language $\mathbf{L_2 \subset L1}$, then **L2** is also Regular. d) If Language L_1 and L_2 are Regular, then $\mathbf{L_1 \cap L_2}$ is also Regular.	4
5	Use Pumping Lemma to comment on the following languages being Regular or NOT: a) L1 = $\{\mathbf{a^p b^q c^r d^s} \mid \mathbf{p+q > r+s}\}$ b) L2 = $\{\mathbf{0^x 1^y 2^z} \mid \mathbf{x, y, z > 2}\}$	4

P.S.: Reasons/Explanation MUST be provided to stated questions, without which no marks shall be given.



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